

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1026

Scrap Disposal Corp 138 kV
July 16, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1026. The Interconnection Customer proposes a 54.9 MW solar photovoltaic facility interconnecting at the Scrap Disposal Corp 138 kV substation in the West Virginia area. The request was received on July 10, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 272 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,401,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Tat Momoli to Grant Town Power Plant 138 KV	West Fork Substation to Rogue Riv 345 KV L	138.4	243	6.5
Milbrae to Chester Generating Sta 138 KV	Halsey to Border Steel 138 KV	118.9	212	13.2
West Fork 138/1kV TX	Harrison to West Fork Substation 345 KV	106.1	435	13.4
Pruntytown to Knoll Trail 138 KV	Rogue River to W Fork 345 KV	101.1	256	6.5
Greenhorn to Pruntytown 138 KV	RogueRiver to West Fork 345 KV	101.1	267	6.5
Greenhorn to Egan 138 KV	West Fork to Rogue Riv 345 KV	101.0	297	6.5
Egan to Tat Momoli 138 KV	Rogue River to W Fork 345 KV	101.0	328	6.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tat Momoli to Grant Town Power Plant 138 KV	Rebuild line with higher-capacity conductor	\$3,488,000	40% (\$1,391,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Milbrae to Chester Generating Sta 138 KV	Rebuild line with higher-capacity conductor	\$15,370,000	43% (\$6,626,000)
West Fork 138-1 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,042,000	20% (\$1,207,000)
Pruntytown to Knoll Trail 138 KV	Reconductor line	\$2,761,000	40% (\$1,101,000)
Greenhorn to Pruntytown 138 KV	Reconductor line	\$1,479,000	40% (\$590,000)
Greenhorn to Egan 138 KV	Reconductor line	\$1,939,000	40% (\$773,000)
Egan to Tat Momoli 138 KV	Reconductor line	\$1,787,000	40% (\$713,000)
	Total allocated to project		\$12,401,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.